

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

|                             |                                                                        |
|-----------------------------|------------------------------------------------------------------------|
| <b>Product Name</b>         | <u>Ethanol, anhydrous, denatured, 90%, 5% methanol, 5% isopropanol</u> |
| <b>Synonyms</b>             | Alcohol; Ethyl alcohol                                                 |
| <b>Molecular Formula</b>    | C <sub>2</sub> H <sub>6</sub> O                                        |
| <b>Molecular Weight</b>     | 46.06                                                                  |
| <b>Recommended Use</b>      | Laboratory chemicals.                                                  |
| <b>Uses advised against</b> | No Information available                                               |

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>22931</b>                                                                                |
| <b>Address</b>                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>09 980 6780 or +64 9 980 6780</b>                                    |
| <b>Telephone / Fax Numbers</b> | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| <b>E-mail address</b>          | <u>ANZinfo@thermofisher.com</u>                                                             |

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

### GHS Classification

#### Physical hazards

Flammable liquids Category 2

#### Health hazards

|                                                      |            |
|------------------------------------------------------|------------|
| Acute Oral Toxicity                                  | Category 4 |
| Acute Inhalation Toxicity - Vapors                   | Category 4 |
| Serious Eye Damage/Eye Irritation                    | Category 2 |
| Reproductive Toxicity                                | Category 2 |
| Specific target organ toxicity - (single exposure)   | Category 2 |
| Specific target organ toxicity - (repeated exposure) | Category 2 |

#### Environmental hazards

Based on available data, the classification criteria are not met

Label Elements



Signal Word

Danger

**Hazard Statements**

H225 - Highly flammable liquid and vapor  
H302 - Harmful if swallowed  
H319 - Causes serious eye irritation  
H332 - Harmful if inhaled  
H371 - May cause damage to organs  
H361 - Suspected of damaging fertility or the unborn child  
H373 - May cause damage to organs through prolonged or repeated exposure

**Precautionary Statements**

**Prevention**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P233 - Keep container tightly closed  
P240 - Ground and bond container and receiving equipment  
P241 - Use explosion-proof electrical/ ventilating/ lighting equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P308 + P311 - IF exposed or concerned: Call a POISON CENTER or doctor  
P312 - Call a POISON CENTER or doctor if you feel unwell  
P330 - Rinse mouth  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

**Storage**

P403 + P235 - Store in a well-ventilated place. Keep cool

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other hazards which do not result in classification**

No information available This product does not contain any known or suspected endocrine disruptors  
Toxic to terrestrial vertebrates

## Section 3 - Composition and Information on Ingredients

| Component         | CAS No  | Weight % |
|-------------------|---------|----------|
| Ethyl alcohol     | 64-17-5 | 90       |
| Methyl alcohol    | 67-56-1 | 5        |
| Isopropyl alcohol | 67-63-0 | 5        |

## Section 4 - First Aid Measures

### Description of first aid measures

|                                            |                                                                                                                                                                                                                                       |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                                                                                                                                                |
| <b>New Zealand Emergency Tel.</b>          | CHEMTREC®<br>09 980 6780 or +64 9 980 6780                                                                                                                                                                                            |
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                                                                                                          |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                                                       |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                                                                                                     |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water.                                                                                                                                                                          |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                      |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                  |
| <b>Most important symptoms and effects</b> | None reasonably foreseeable. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting; Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting |
| <b>Notes to Physician</b>                  | Treat symptomatically. Symptoms may be delayed.                                                                                                                                                                                       |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam. Water mist may be used to cool closed containers.

### Extinguishing media which must not be used for safety reasons

No information available.

### Specific Hazards Arising from the Chemical

Flammable. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated. Vapors may form explosive mixtures with air.

### Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

### Personal Precautions, Protective Equipment and Emergency Procedures

#### Emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.

#### Environmental Precautions

Should not be released into the environment.

#### Methods for Containment and Clean Up

Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

#### Precautions to prevent secondary hazards

Clean contaminated objects and areas thoroughly observing environmental regulations

#### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

#### Precautions for Safe Handling

##### Advice on safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

##### Hygiene Measures

When using do not eat, drink or smoke. Provide regular cleaning of equipment, work area and clothing.

#### Conditions for Safe Storage, Including any Incompatibilities

##### Storage Conditions

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep away from heat, sparks and flame. Flammables area. Keep container tightly closed in a dry and well-ventilated place.

##### Incompatible Materials

Acids. Ammonia. Peroxides. Acid anhydrides. Acid chlorides. Metals. Reducing Agent.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## Section 8 - Exposure Controls and Personal Protection

#### Control parameters

##### Exposure limits

**NZ** - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

**UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

| Component         | New Zealand WEL                                                                                    | Australia                                                                                   | ACGIH TLV                             | The United Kingdom                                                                                                  |
|-------------------|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Ethyl alcohol     | TWA: 1000 ppm<br>TWA: 1880 mg/m <sup>3</sup>                                                       | TWA: 1000 ppm<br>TWA: 1880 mg/m <sup>3</sup>                                                | STEL: 1000 ppm                        | TWA: 1000 ppm TWA; 1920 mg/m <sup>3</sup> TWA<br>WEL - STEL: 3000 ppm<br>STEL; 5760 mg/m <sup>3</sup> STEL          |
| Methyl alcohol    | TWA: 200 ppm<br>TWA: 262 mg/m <sup>3</sup><br>STEL: 250 ppm<br>STEL: 328 mg/m <sup>3</sup><br>Skin | STEL: 250 ppm<br>STEL: 328 mg/m <sup>3</sup><br>TWA: 200 ppm<br>TWA: 262 mg/m <sup>3</sup>  | TWA: 200 ppm<br>STEL: 250 ppm<br>Skin | WEL - TWA: 200 ppm TWA;<br>266 mg/m <sup>3</sup> TWA<br>WEL - STEL: 250 ppm<br>STEL; 333 mg/m <sup>3</sup> STEL     |
| Isopropyl alcohol | TWA: 400 ppm<br>TWA: 983 mg/m <sup>3</sup><br>STEL: 500 ppm<br>STEL: 1230 mg/m <sup>3</sup>        | STEL: 500 ppm<br>STEL: 1230 mg/m <sup>3</sup><br>TWA: 400 ppm<br>TWA: 983 mg/m <sup>3</sup> | TWA: 200 ppm<br>STEL: 400 ppm         | STEL: 500 ppm 15 min<br>STEL: 1250 mg/m <sup>3</sup> 15 min<br>TWA: 400 ppm 8 hr<br>TWA: 999 mg/m <sup>3</sup> 8 hr |

### Biological limit values

**NZ** - Substances assigned Biological Exposure Indices in the New Zealand Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**ACGIH** - American Conference of Governmental Industrial Hygienists (ACGIH) TLVs® and BEIs®- Threshold Limit Values for Chemical Substances and Physical Agents & Biological Exposure Indices. 2022 Edition

| Component         | New Zealand                                      | Australia | ACGIH - Biological Exposure Indices                                                          | United Kingdom |
|-------------------|--------------------------------------------------|-----------|----------------------------------------------------------------------------------------------|----------------|
| Methyl alcohol    | 15 mg/L (urine) end of shift<br>(Methyl alcohol) |           | 15 mg/L<br>Medium: urine<br>Time: end of shift<br>Determinant: Methanol                      |                |
| Isopropyl alcohol |                                                  |           | 40 mg/L<br>Medium: urine<br>Time: end of shift at end of<br>workweek<br>Determinant: Acetone |                |

### Appropriate engineering controls

#### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Individual protection measures, such as personal protective equipment

**Eye Protection** Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection** Protective gloves

| Glove material          | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments        |
|-------------------------|-------------------|-----------------|-----------------|-----------------------|
| Butyl rubber, Neoprene, | > 480 minutes     | > 0.3 mm        | AS/NZS 2161     | (minimum requirement) |
| Viton (R).              | > 480 minutes     |                 |                 |                       |
|                         | > 480 minutes     |                 |                 |                       |
| Nitrile rubber          | < 60 minutes      |                 |                 |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection** Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387 (or AUS/NZ equivalent)

**Recommended half mask:-** Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** When using do not eat, drink or smoke. Provide regular cleaning of equipment, work area and clothing.

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                                             |                                          |
|------------------------------------------------|---------------------------------------------|------------------------------------------|
| <b>Physical State</b>                          | Liquid                                      |                                          |
| <b>Appearance</b>                              | Colorless                                   |                                          |
| <b>Odor</b>                                    | No information available                    |                                          |
| <b>Odor Threshold</b>                          | No data available                           |                                          |
| <b>pH</b>                                      | Not applicable                              |                                          |
| <b>Melting Point/Range</b>                     | -114 °C / -173.2 °F                         |                                          |
| <b>Softening Point</b>                         | No data available                           |                                          |
| <b>Boiling Point/Range</b>                     | 78 °C / 172.4 °F                            | @ 760 mmHg                               |
| <b>Flammability (liquid)</b>                   | Highly flammable                            | On basis of test data                    |
| <b>Flammability (solid,gas)</b>                | Not applicable                              | Liquid                                   |
| <b>Explosion Limits</b>                        | <b>Lower</b> 4 vol%<br><b>Upper</b> 19 vol% |                                          |
| <b>Flash Point</b>                             | 12 °C / 53.6 °F                             | <b>Method</b> - No information available |
| <b>Autoignition Temperature</b>                | 370 °C / 698 °F                             |                                          |
| <b>Decomposition Temperature</b>               | No data available                           |                                          |
| <b>Viscosity</b>                               | 1.2 mPa.s at 20 °C                          |                                          |
| <b>Water Solubility</b>                        | Miscible                                    |                                          |
| <b>Solubility in other solvents</b>            | No information available                    |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                             |                                          |
| <b>Component</b>                               | <b>log Pow</b>                              |                                          |
| Ethyl alcohol                                  | -0.32                                       |                                          |
| Methyl alcohol                                 | -0.74                                       |                                          |
| Isopropyl alcohol                              | 0.05                                        |                                          |
| <b>Vapor Pressure</b>                          | 59 mbar @ 20 °C                             |                                          |
| <b>Density / Specific Gravity</b>              | 0.780                                       |                                          |
| <b>Bulk Density</b>                            | Not applicable                              | Liquid                                   |
| <b>Vapor Density</b>                           | 1.59                                        | (Air = 1.0)                              |
| <b>Particle characteristics</b>                | Not applicable (liquid)                     |                                          |

### Other information

|                             |                                             |
|-----------------------------|---------------------------------------------|
| <b>Molecular Formula</b>    | C2 H6 O                                     |
| <b>Molecular Weight</b>     | 46.06                                       |
| <b>Explosive Properties</b> | Vapors may form explosive mixtures with air |

## Section 10 - Stability and Reactivity

|                                         |                                                                                          |
|-----------------------------------------|------------------------------------------------------------------------------------------|
| <b>Reactivity</b>                       | None known, based on information available                                               |
| <b>Stability</b>                        | Stable under normal conditions.                                                          |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                 |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                 |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                 |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                            |
| <b>Conditions to Avoid</b>              | Keep away from open flames, hot surfaces and sources of ignition, Incompatible products. |
| <b>Incompatible Materials</b>           | Acids, Ammonia, Peroxides, Acid anhydrides, Acid chlorides, Metals, Reducing Agent.      |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ).                                 |

## Section 11 - Toxicological Information

### Acute Effects

#### Information on likely routes of exposure

#### Product Information

|                   |                                                                                                                  |
|-------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b> | Avoid breathing vapors or mists. Harmful by inhalation.                                                          |
| <b>Eyes</b>       | Avoid contact with eyes. Irritating to eyes. Vapor may cause irritation.                                         |
| <b>Skin</b>       | Avoid contact with skin. May cause irritation. Prolonged skin contact may defat the skin and produce dermatitis. |
| <b>Ingestion</b>  | May be harmful if swallowed.                                                                                     |

#### Numerical measures of toxicity

##### (a) acute toxicity;

|                   |                                                                  |
|-------------------|------------------------------------------------------------------|
| <b>Oral</b>       | Category 4                                                       |
| <b>Dermal</b>     | Based on available data, the classification criteria are not met |
| <b>Inhalation</b> | Category 4                                                       |

#### Toxicology data for the components

| Component         | LD50 Oral                                                    | LD50 Dermal                   | LC50 Inhalation                                                   |
|-------------------|--------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------|
| Ethyl alcohol     | LD50 = 10470 mg/kg<br>OECD 401 (Rat)<br>3450 mg/kg ( Mouse ) |                               | LC50 = 117-125 mg/l (4h)<br>OECD 403 (rat)<br>20000 ppm/10H (rat) |
| Methyl alcohol    | LD50 = 1187 – 2769 mg/kg (Rat)                               | LD50 = 17100 mg/kg ( Rabbit ) | LC50 = 128.2 mg/L ( Rat ) 4 h                                     |
| Isopropyl alcohol | 5045 mg/kg ( Rat )<br>3600 mg/kg ( Mouse )                   | 12800 mg/kg ( Rat )           | 72.6 mg/L ( Rat ) 4 h                                             |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; Category 2

##### (d) respiratory or skin sensitization;

|                    |                   |
|--------------------|-------------------|
| <b>Respiratory</b> | No data available |
| <b>Skin</b>        | No data available |

| Component                       | Test method                                                    | Test species | Study result    |
|---------------------------------|----------------------------------------------------------------|--------------|-----------------|
| Ethyl alcohol<br>64-17-5 ( 90 ) | Mouse Ear Swelling Test (MEST)                                 | mouse        | non-sensitising |
|                                 | OECD Test Guideline 429 Local Lymph Node Assay                 | mouse        | non-sensitising |
| Methyl alcohol<br>67-56-1 ( 5 ) | OECD Test Guideline 406<br>Guinea Pig Maximisation Test (GPMT) | guinea pig   | non-sensitising |

(e) germ cell mutagenicity; No data available

| Component                       | Test method                                   | Test species          | Study result |
|---------------------------------|-----------------------------------------------|-----------------------|--------------|
| Ethyl alcohol<br>64-17-5 ( 90 ) | AMES test<br>OECD Test Guideline 471          | in vitro<br>Bacteria  | negative     |
|                                 | Gene cell mutation<br>OECD Test Guideline 476 | in vitro<br>Mammalian | negative     |

(f) carcinogenicity; No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

**(g) reproductive toxicity;**

No data available

| Component                       | Test method             | Test species / Duration       | Study result           |
|---------------------------------|-------------------------|-------------------------------|------------------------|
| Ethyl alcohol<br>64-17-5 ( 90 ) | OECD Test Guideline 416 | Oral / mouse 2 Generation     | NOAEL = 13.8 g/kg/day  |
|                                 | OECD Test Guideline 414 | Inhalation / Rat              | NOAEC = 16000 ppm      |
| Methyl alcohol<br>67-56-1 ( 5 ) | OECD Test Guideline 416 | Rat / Inhalation 2 Generation | NOAEC = 1.3 mg/l (air) |

**Reproductive Effects**

California Proposition 65 Reproductive toxicity

**(h) STOT-single exposure;**

Category 2

**Results / Target organs**Optic nerve  
Central nervous system (CNS)**(i) STOT-repeated exposure;**

No data available

**Target Organs**

None known.

**(j) aspiration hazard;**

No data available

**Symptoms / effects, both acute and delayed**

Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

## Section 12 - Ecological Information

**Ecotoxicity****Aquatic ecotoxicity**

Contains a substance which is: Toxic to aquatic organisms. The product contains following substances which are hazardous for the environment.

| Component         | Freshwater Fish                                                                                                                                                                                                                   | Water Flea                                         | Freshwater Algae                                                                                                 | Microtox                                                                                                                |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Ethyl alcohol     | Fathead minnow<br>(Pimephales promelas)<br>LC50 = 14200 mg/l/96h                                                                                                                                                                  | EC50 = 9268 mg/L/48h<br>EC50 = 10800 mg/L/24h      | EC50 (72h) = 275 mg/l<br>(Chlorella vulgaris)                                                                    | Photobacterium<br>phosphoreum: EC50 =<br>34634 mg/L/30 min<br>Photobacterium<br>phosphoreum: EC50 =<br>35470 mg/L/5 min |
| Methyl alcohol    | Pimephales promelas:<br>LC50 > 10000 mg/L 96h                                                                                                                                                                                     | EC50 > 10000 mg/L 24h                              |                                                                                                                  | EC50 = 39000 mg/L 25<br>min<br>EC50 = 40000 mg/L 15<br>min<br>EC50 = 43000 mg/L 5<br>min                                |
| Isopropyl alcohol | LC50: = 9640 mg/L, 96h<br>flow-through<br>(Pimephales promelas)<br>LC50: > 1400000 µg/L,<br>96h (Lepomis<br>macrochirus)<br>LC50: = 11130 mg/L,<br>96h static (Pimephales<br>promelas)<br>LC50: = 10000000 µg/L,<br>96h (Daphnia) | 13299 mg/L EC50 = 48<br>h<br>9714 mg/L EC50 = 24 h | EC50: > 1000 mg/L, 72h<br>(Desmodesmus<br>subspicatus)<br>EC50: > 1000 mg/L, 96h<br>(Desmodesmus<br>subspicatus) | = 35390 mg/L EC50<br>Photobacterium<br>phosphoreum 5 min                                                                |

**Terrestrial ecotoxicity**

| Component     | Earthworm                                                                                   | Avian | Honeybees |
|---------------|---------------------------------------------------------------------------------------------|-------|-----------|
| Ethyl alcohol | Acute toxicity: LC50 0.1 - 1<br>mg/cm <sup>2</sup> (Eisenia foetida, 48 h,<br>filter paper) |       |           |

|                |                                                                                      |  |  |
|----------------|--------------------------------------------------------------------------------------|--|--|
| Methyl alcohol | Acute toxicity: LC50 > 1 mg/cm <sup>2</sup><br>(Eisenia foetida, 48 h, filter paper) |  |  |
|----------------|--------------------------------------------------------------------------------------|--|--|

**Persistence and Degradability** Expected to be biodegradable

**Persistence** Persistence is unlikely, based on information available.

| Component                       | Degradability                  |
|---------------------------------|--------------------------------|
| Ethyl alcohol<br>64-17-5 ( 90 ) | OECD 301E = 94%                |
| Methyl alcohol<br>67-56-1 ( 5 ) | DT50 ~ 17.2d<br>>94% after 20d |

**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential** Bioaccumulation is unlikely

| Component         | log Pow | Bioconcentration factor (BCF) |
|-------------------|---------|-------------------------------|
| Ethyl alcohol     | -0.32   | No data available             |
| Methyl alcohol    | -0.74   | <10 dimensionless             |
| Isopropyl alcohol | 0.05    | No data available             |

**Mobility** The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility. Disperses rapidly in air

#### Other adverse effects

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

#### Waste treatment methods

**Waste from Residues/Unused Products** Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**Other Information** Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations . Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations.

## Section 14 - Transport Information

| Component                       | Hazchem Code |
|---------------------------------|--------------|
| Ethyl alcohol<br>64-17-5 ( 90 ) | 2YE<br>2Y    |
| Methyl alcohol<br>67-56-1 ( 5 ) | 2WE          |

|                                    |    |
|------------------------------------|----|
| Isopropyl alcohol<br>67-63-0 ( 5 ) | 1Z |
|------------------------------------|----|

**NZS 5433:2020**

|                         |                              |
|-------------------------|------------------------------|
| UN-No                   | UN1987                       |
| Proper Shipping Name    | ALCOHOLS, N.O.S.             |
| Technical Shipping Name | Ethanol/Methanol/Isopropanol |
| Hazard Class            | 3                            |
| Packing Group           | II                           |

**IATA**

|                         |                              |
|-------------------------|------------------------------|
| UN-No                   | UN1987                       |
| Proper Shipping Name    | ALCOHOLS, N.O.S.             |
| Technical Shipping Name | Ethanol/Methanol/Isopropanol |
| Hazard Class            | 3                            |
| Packing Group           | II                           |

**IMDG/IMO**

|                         |                              |
|-------------------------|------------------------------|
| UN-No                   | UN1987                       |
| Proper Shipping Name    | ALCOHOLS, N.O.S.             |
| Technical Shipping Name | Ethanol/Methanol/Isopropanol |
| Hazard Class            | 3                            |
| Packing Group           | II                           |

**Environmental hazards** No hazards identified

**Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code** Not applicable, packaged goods

**Special Precautions** No special precautions required. Please refer to the applicable dangerous goods regulations for additional information.

**Additional information** None known

## **Section 15 - Regulatory Information**

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

**National Regulations**

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

**Certified handlers, tracking and controlled substance license requirements**

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

| Component      | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | IMDG Marine Pollutant |
|----------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-----------------------|
| Methyl alcohol | 500 tonne                                                                                 | 5000 tonne                                                                               |                       |

#### Authorisation/Restrictions according to EU REACH

| Component         | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                      | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Methyl alcohol    | -                                                                   | Use restricted. See item 69. (see link for restriction details)<br>Use restricted. See item 75. (see link for restriction details) | -                                                                                                     |
| Isopropyl alcohol | -                                                                   | Use restricted. See item 75. (see link for restriction details)                                                                    | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

#### International Inventories

China, X = listed, Australia, U.S.A. (TSCA), Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Australia (AICS), Korea (KECL), China (IECSC), Japan (ENCS), Philippines (PICCS), Taiwan (TCSI), Japan (ISHL), New Zealand (NZIoC), Japan (ISHL). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component         | CAS No  | NZIoC | AICS | EINECS | ELINCS | NLP | KECL     | IECSC | TCSI |
|-------------------|---------|-------|------|--------|--------|-----|----------|-------|------|
| Ethyl alcohol     | 64-17-5 | X     | X    | -      | -      | -   | KE-13217 | X     | X    |
| Methyl alcohol    | 67-56-1 | X     | X    | -      | -      | -   | KE-23193 | X     | X    |
| Isopropyl alcohol | 67-63-0 | X     | X    | -      | -      | -   | KE-29363 | X     | X    |

| Component         | CAS No  | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | PICCS | ISHL | ENCS |
|-------------------|---------|------|-----------------------------------------------|-----|------|-------|------|------|
| Ethyl alcohol     | 64-17-5 | X    | ACTIVE                                        | X   | -    | X     | X    | X    |
| Methyl alcohol    | 67-56-1 | X    | ACTIVE                                        | X   | -    | X     | X    | X    |
| Isopropyl alcohol | 67-63-0 | X    | ACTIVE                                        | X   | -    | X     | X    | X    |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Section 16 - Other Information

**This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations**

#### Legend

**NZIoC** - New Zealand Inventory of Chemicals

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**IECSC** - Chinese Inventory of Existing Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**NZS 5433:2020** - Transport of Dangerous Goods on Land

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**AICS** - Australian Inventory of Chemical Substances

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**CAS** - Chemical Abstracts Service

**ACGIH** - American Conference of Governmental Industrial Hygienists

**PNEC** - Predicted No Effect Concentration

**OECD** - Organisation for Economic Co-operation and Development

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**WEL** - Workplace Exposure Limit

**DNEL** - Derived No Effect Level

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**VOC** - (Volatile Organic Compound)

**LC50** - Lethal Concentration 50%

**ATE** - Acute Toxicity Estimate

**RPE** - Respiratory Protective Equipment

**NOEC** - No Observed Effect Concentration

**BCF** - Bioconcentration factor

**PBT** - Persistent, Bioaccumulative, Toxic

#### Key literature references and sources for data

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

EPA Guide to classifying hazardous substances in New Zealand

EPA - Assigning a product to an existing HSNO approval guide

#### Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

#### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

**Revision Date** 20-Mar-2023

**Revision Summary** SDS sections updated

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**